

Towards Transfer Fabrication of Alumina-Epoxy Acoustic Matching Layers for PZT Ultrasound Transducers

Tomasz Zaluska^{1,3}, Christoph Leitner², Younghun Kim⁴, Federico Villani², Butrus Khuri-Yakub⁴, Luca Benini², Todd Coleman¹, Zhenan Bao³

¹Department of Bioengineering, Stanford University, Stanford, USA

²Integrated Systems Laboratory, ETH Zurich, Zurich, Switzerland

³Department of Chemical Engineering, Stanford University, Stanford, USA

⁴Department of Electrical Engineering, Stanford University, Stanford, USA

Background, Motivation and Objective

Efficient ultrasound transmission from high-impedance piezoelectric ceramics into water or soft tissue requires acoustic matching layers with controlled acoustic impedance, quarter-wavelength thickness, and low attenuation. For a single matching layer between PZT and a water/tissue load, the target impedance is approximately the geometric mean of the ceramic and load impedances, giving a practical design range of approximately 5-8 MRayl. Conventional transfer-based fabrication approaches can introduce acoustic discontinuities that reduce the efficiency of energy transfer from the PZT into the propagation medium. To address this limitation, we develop and characterize an alumina–epoxy composite matching layer whose acoustic impedance can be tuned through the alumina filler loading. We demonstrate the integration of this matching-layer structure with a transfer-compatible fabrication process for MHz PZT ultrasound devices, enabling scalable device fabrication and system-level integration, including implementation on silicon carrier wafers.

Statement of Contribution/Methods

Alumina–epoxy bulk coupons were fabricated by loading epoxy (Epotek 301) with alumina at the volume fractions shown in Fig. 1, casting into spacer-defined molds, curing under compression, and releasing planar samples. Acoustic characterization was performed on $N = 8$ coupons with 0, 10, 19, and 25 vol% alumina loading. Longitudinal sound speed and normal-incidence reflection delay were extracted from front- and back-wall echo time-of-flight in a water-tank pulse-echo setup. Attenuation was estimated from the frequency-domain amplitude loss between echoes over the round-trip path. Impedance was calculated using the measured density and speed of sound ($Z = \rho \cdot c$). To implement the transfer-compatible fabrication process, the selected alumina–epoxy formulation was cast on a dextran-coated silicon carrier wafer, cured, polished to the 2 MHz quarter-wavelength target thickness, bonded to PZT with a thin adhesive layer ($\approx 1 \mu\text{m}$), post-cured, and released by dissolving the dextran in water.

Results/Discussion

Alumina loading increased the acoustic impedance of the epoxy composite from approximately 3.1 MRayl for unfilled epoxy to 6.0 MRayl at 25 vol% alumina, approaching the impedance range required for single-layer matching between PZT and water or soft tissue. The 19–25 vol% formulations therefore define the most relevant processing window for device integration. Importantly, these compositions remained compatible with casting into thin planar films, polishing to a quarter-wavelength thickness, bonding to PZT with a thin adhesive layer, and release from the sacrificial carrier. This links the material optimization directly to a transfer-compatible fabrication route for compact, integrated MHz ultrasound transducers.

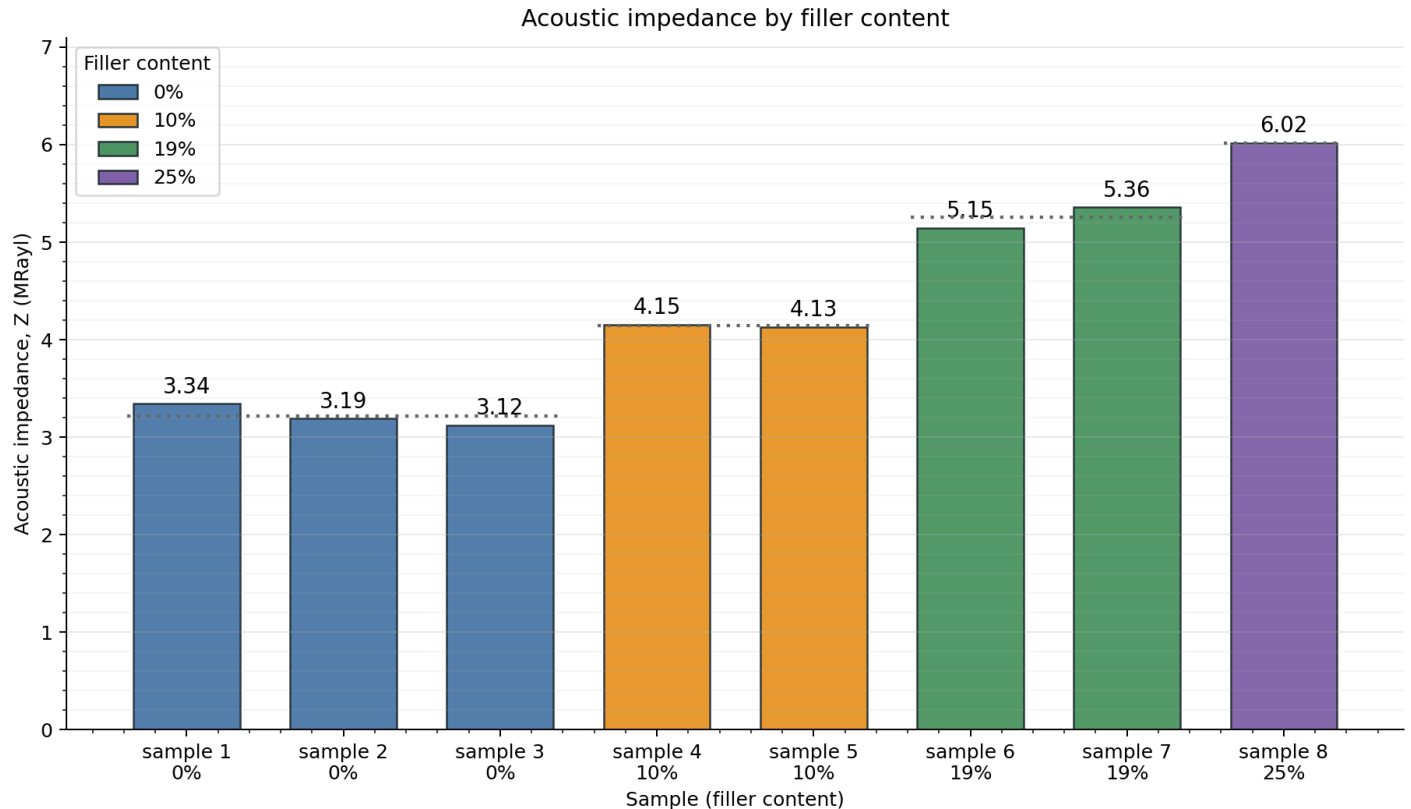


Figure 1. Acoustic impedance of alumina-epoxy coupons as a function of alumina filler content. Values were obtained by reflection-delay analysis using measured sample thicknesses. The data show a monotonic impedance increase with filler content and support selection of the highest-loaded processable formulations for subsequent matching-layer fabrication.